HARI SANKAR KELUTH

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SUMMARY

AI Engineer with 4+ years of experience delivering **Traditional ML and Generative AI solutions at enterprise scale**. Specialized in **Agentic AI, Retrieval-Augmented Generation (RAG), prompt engineering, MCP, and MLOps**, with a strong track record of building LLM-powered applications, document generation systems, and multi-agent workflows that drive measurable business impact. Experienced in designing end-to-end AI pipelines, scalable APIs, and cloud-native AI systems on **AWS and GCP**, with deep expertise in integrating structured and unstructured data for intelligent automation.

EXPERIENCE

Grantide – Remote, USA **January 2025 – Present**

AI Engineer *[Python, LangChain, LangGraph, RAG, Agentic AI, Prompt Engineering, FAISS, AWS, GCP, FastAPI, Firestore, Docker]*

- Engineered a **RAG-based AI platform** for automated grant discovery and proposal generation, enabling organizations to identify funding opportunities and produce funder-ready documents in minutes, reducing manual proposal effort by ~60%.
- Designed **document intelligence pipelines** that extracted organizational and project-level context to generate structured grant proposals precisely tailored to funder requirements.
- Built **semantic search and retrieval systems** using FAISS and vector embeddings, achieving ~40% improvement in grant-opportunity matching accuracy for client organizations.
- Developed **end-to-end LLM pipelines** with advanced prompt engineering and structured outputs, significantly reducing hallucinations and ensuring factual grounding in generated proposals.
- Designed and deployed **scalable backend APIs** (FastAPI/Flask) on AWS (EC2, S3, Lambda) and GCP Cloud Run, supporting multi-tenant AI workflows with real-time inference at scale.
- Implemented **analytics and monitoring pipelines** to track proposal success rates and system performance metrics, directly improving deployment reliability and business decision-making.
- Containerized applications using Docker and established **CI/CD pipelines**, enabling repeatable deployments and reducing production downtime by ~30%.
- Accelerated proposal generation speed by **~3x** through AI-powered automation workflows, collaborating cross-functionally with product and domain teams.

Cadential Technologies – India **March 2021 – July 2023**

Machine Learning Engineer *[Python, Machine Learning, PySpark, Elasticsearch, AWS, Docker, Kubernetes, SQL, Kafka]*

- Developed scalable **machine learning pipelines** and backend systems for large-scale data processing, supporting analytics applications serving millions of records.
- Built **semantic search** and retrieval systems using Elasticsearch, improving query relevance and reducing end-user response latency by ~35%.
- Designed and implemented **ETL pipelines** using PySpark, processing multi-million-record datasets efficiently for downstream ML model consumption.
- Deployed ML services on **AWS infrastructure** (EC2, S3, Lambda) with Docker-based containerization, enhancing scalability, fault tolerance, and operational reliability.
- Integrated distributed messaging pipelines using **Apache Kafka**, enabling robust real-time data ingestion and improving downstream processing throughput.
- Delivered production-grade AI solutions in Agile environments through close collaboration with cross-functional engineering and product teams.

AICTE – Remote **October 2021 – December 2021**

AWS Cloud Intern *[AWS (S3, Glue, CloudWatch), SQL, Python]*

- Built **ETL automation workflows** using AWS Glue and S3, enabling repeatable and scalable data processing pipelines for downstream analytics.
- Developed **SQL-based validation** and analytics workflows to ensure data quality, consistency, and integrity across ingested datasets.
- Implemented **CloudWatch monitoring** and alerting systems, improving pipeline reliability and providing real-time operational visibility into data workflows.
- Optimized **S3 storage** configurations and ETL processes, reducing redundant processing overhead and improving overall pipeline performance efficiency.

SKILLS

- **Programming:** Python, SQL, Java, JavaScript, TypeScript, Bash, C++, C, R
- **Generative AI & Agentic AI:** Agentic AI, Multi-Agent Systems, RAG, Prompt Engineering, Tool Calling, Function Calling, ReAct Agents, MCP (Model Context Protocol), LLM Fine-Tuning (LoRA/QLoRA), Multimodal AI
- **LLMs & Platforms:** GPT-4, Claude, Gemini, LLaMA, Mixtral, ChatGroq, OpenAI API, Hugging Face
- **Frameworks & AI Tooling:** LangChain, LangGraph, LlamaIndex, LangSmith, FAISS, ChromaDB, AutoGen, CrewAI, AutoGPT, LangFlow, Agno, FastAPI, Flask, Streamlit, Pydantic
- **Vector Databases & Retrieval:** FAISS, Pinecone, Weaviate, Elasticsearch, OpenSearch, Embeddings, Semantic & Hybrid Search
- **Backend & APIs:** FastAPI, Flask, REST APIs, Microservices, Async Workflows
- **Cloud & MLOps:** AWS (EC2, S3, Lambda, ECS, EKS, VPC, CloudFront, API Gateway, SageMaker, Bedrock, CloudWatch), GCP (Cloud Run, BigQuery), Azure (OpenAI, Document Intelligence), Docker, Kubernetes, CI/CD, MLflow, Data Version Control
- **Databases & Data Engineering:** PostgreSQL, MySQL, MongoDB, Firebase Firestore, BigQuery, ETL Pipelines, PySpark, Kafka
- **Tools & Platforms:** GitHub, GitHub Actions, Jupyter Notebook, n8n, Tableau, Power BI, Microsoft Copilot, Copilot Studio

EDUCATION

University of North Texas, M.S. in Computer & Information Science **August 2023 – May 2025**

Coursework: Machine Learning, Cloud Computing, Artificial Intelligence, Deep Learning, Distributed Systems

VR Siddhartha Engineering College, B.Tech in Computer Science Engineering **August 2019 – May 2023**

Coursework: Data Structures, Algorithms, Database Management, Operating Systems, Artificial Intelligence

PROJECTS

Multi-Agent AI Orchestration System *[LangGraph, LangChain, Multi-Agent Systems, Prompt Engineering, Tool Calling]*

- Engineered a **multi-agent architecture** comprising Supervisor, Researcher, Enhancer, Coder, and Validator agents, enabling dynamic task routing and intelligent decision-making across complex workflows.
- Designed **agent orchestration pipelines** with iterative reasoning and validation loops, improving response accuracy by ~35% and ensuring output reliability across diverse task types.
- Integrated **real-time tool execution** (web search, code interpreter) enabling agents to retrieve live information and solve dynamic problems autonomously.

Full-Stack AI Agent Chatbot *[FastAPI, LangGraph, LangChain, RAG, Tavily Search, Docker, Streamlit, LLMs]*

- Built a **production-ready**, full-stack AI agent chatbot with a FastAPI backend and Streamlit UI, supporting real-time multi-turn interactions and AI-driven dynamic responses.
- Implemented **ReAct-based agent workflows** with integrated tool use (Tavily Search), enabling multi-step reasoning and live external information retrieval.
- Supported **dynamic LLM selection** (LLaMA, Mixtral) and customizable system prompts, providing flexible deployment scenarios; Dockerized for production-ready scalability.

RAG-Based Intelligent Document Chatbot *[LangChain, FAISS, Embeddings, NLP, Python]*

- Developed a document-aware chatbot using end-to-end **RAG pipelines**, enabling context-precise question answering from PDFs and unstructured text data.
- Implemented **chunking, vector embedding**, and semantic retrieval pipelines with FAISS, reducing manual information search effort by ~40% and improving answer relevance.

LLM Fine-Tuning & RAG Optimization *[Hugging Face, LoRA/QLoRA, Transformers, NLP]*

- Fine-tuned open-source LLMs (LLaMA, Mistral) using **LoRA/QLoRA** parameter-efficient techniques, achieving significant improvements in domain-specific response quality and contextual coherence.
- Integrated fine-tuned models into **RAG-based production pipelines** with full evaluation workflows, demonstrating measurable gains over baseline retrieval-only approaches.

CERTIFICATIONS

- Google Professional Machine Learning Engineer – Google Cloud
- AWS Solutions Architect – Associate – Amazon Web Services
- AWS Cloud Foundations – Amazon Web Services